Exercice 1

a) Trois facteurs négatifs : le résultat est négatif. $(-2) \times (-5) \times (-3) = \mathbf{-30}$

b) On calcule de gauche à droite. $(-36) \div (-4) \times (-2) = 9 \times (-2) = \mathbf{-18}$

c) $\dfrac{(-3) \times (-10)}{-6} = \dfrac{30}{-6} = \mathbf{-5}$

Exercice 2

a) $(-3)^2 = (-3) \times (-3) = \mathbf{9}$

b) $-3^2 = -(3 \times 3) = \mathbf{-9}$: le carré ne porte que sur 3.

c) $(-2)^3 = (-2) \times (-2) \times (-2) = \mathbf{-8}$

d) 15 est impair, donc $(-1)^{15} = \mathbf{-1}$.

Exercice 3

$A = 5 - 3 \times \mathbf{(-2)^2}$
$A = 5 - 3 \times 4$
$A = 5 - 12$
$A = \mathbf{-7}$

$C = \dfrac{-2^3 + 4 \times (-5)}{(-7) \times 2^2}$
$C = \dfrac{-8 + (-20)}{(-7) \times 4}$
$C = \dfrac{-28}{-28}$
$C = \mathbf{1}$

Exercice 4

a) $(-5)^{-2} = \dfrac{1}{(-5)^2} = \mathbf{\dfrac{1}{25}}$

b) $5^0 + 5^{-1} = 1 + \dfrac{1}{5} = \dfrac{5}{5} + \dfrac{1}{5} = \mathbf{\dfrac{6}{5}}$ (attention : $5^0 = 1$, et non 0)

c) $\left(\dfrac{2}{3}\right)^{-2} = \left(\dfrac{3}{2}\right)^{2} = \mathbf{\dfrac{9}{4}}$

Exercice 5

a) $\dfrac{5^3 \times 5^{-7}}{5^{-2}} = \dfrac{5^{3+(-7)}}{5^{-2}} = \dfrac{5^{-4}}{5^{-2}} = 5^{-4-(-2)} = \mathbf{5^{-2}}$

b) $\dfrac{(7^2)^3}{7^4 \times 7^{-1}} = \dfrac{7^6}{7^3} = 7^{6-3} = \mathbf{7^3}$

c) $9 = 3^2$ donc $9^{-3} = (3^2)^{-3} = 3^{-6}$. Ainsi $3^4 \times 9^{-3} = 3^4 \times 3^{-6} = \mathbf{3^{-2}}$

d) $2^5 \times 5^5 = (2 \times 5)^5 = \mathbf{10^5}$

Exercice 6

1. 1 cm = 0,01 m = $\mathbf{10^{-2}}$ m.

2. $\dfrac{10^{-2}}{2 \times 10^{-5}} = \dfrac{1}{2} \times 10^{-2-(-5)} = 0{,}5 \times 10^3 = 500$. Il faut aligner **500 cellules**.